

Möbius-Weighted Shared-Frequency Clusters in the Physical Cross-Divisor Gram

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Abstract

TPC-213 showed that the literal transition residual is generated by one common physical source and that its divisor cross terms are indexed by shared rational frequencies. This paper adds the actual Möbius-log coefficient to that frequency picture. For a common reciprocal prime set, common height, and smooth emitter, we prove an exact dilation covariance: if $h \mid d$ then the row at denominator d restricted to frequencies with reduced denominator h is exactly the row at h . Consequently the complete-period physical Gram factors through reduced-denominator clusters, with cluster coefficient $C_h = \sum_{h \mid d} \mu(d) \log(d)/d$. The zero axis vanishes when the prime shell lies below the Fourier height, and the same reduction commutes with the four-packet polarization used in the Gate-B route. Exact rational row data, followed by numerical logarithm evaluation, gives a weighted physical/direct-sum ratio 0.5963435557 for $\{5, 7, 35\}$ and 1.2119952513 for $\{3, 5, 7, 105\}$. Thus the cluster reduction is a genuine analytic structure, but it has no universal favorable sign. No asymptotic saving, prime-shell reassembly, arithmetic L^2 credit, or twin-prime conclusion is claimed.

Claim level. PROVED_STRUCTURAL_L1 / MOBIUS_CLUSTER_REDUCTION.

The finite sign directions are exact; the displayed ratios are numerical observations reconstructed from exact rational rows; the literal V46 asymptotic cluster bound remains open.

1 Introduction

The transition-native twin-prime route contains one reciprocal emitter row for each squarefree divisor. In the V46 normalization the row has the schematic form

$$A_d(r) = \frac{\mu(d) \log d}{d} \sum_{q \in \mathcal{Q}} \sum_{0 < |m| \leq dq/H} \psi\left(\frac{Hm}{dq}\right) \mathbf{1}_{r \equiv m\bar{q} \pmod{d}}. \quad (1)$$

This is the transition-native object frozen in the V46 research artifact [1]. TPC-212 isolated its truncated emitter boundary [3], and TPC-213 supplied the common-source pullback and the pairwise frequency-intersection Gram [2]. The common-source pullback from TPC-213 showed that the corresponding physical kernel is not an orthogonal direct sum. On a complete period, only equal rational frequencies interact. The remaining question is whether the Möbius-log coefficients make those interactions cancel in a systematic way.

The answer has two parts. First, there is an exact algebraic collapse. Strip the coefficient from (1) and call the resulting row B_d . Whenever $h \mid d$, the frequency $r = (d/h)a$ has the same amplitude

as the reduced row $B_h(a)$. All divisors that are multiples of h therefore share one row, and their coefficients add before squaring. This is stronger than a list of pairwise Gram entries: it identifies the correct reduced object.

Second, the coefficient tail is not sign-definite. The nested family $\{5, 7, 35\}$ produces a substantial finite reduction, while the family $\{3, 5, 7, 105\}$ produces a finite enhancement. The examples do not model the full asymptotic problem, but they rule out a universal theorem whose only input is the existence of shared-frequency coupling.

The contributions are therefore:

1. an exact dilation covariance for the literal reciprocal cutoff;
2. an exact reduced-denominator factorization of the common-source Gram;
3. an exact zero-axis and four-packet compatibility statement;
4. two independently checked finite coefficient ledgers with opposite qualitative behavior.

2 The literal emitter and its reduced frequencies

Let $\mathcal{D}$ be a finite family of squarefree positive integers. Let $\mathcal{Q}$ be a finite set of integers coprime to every member of $\mathcal{D}$, let $H \geq 1$, and let ψ be any function on the finite set of arguments that occur below. Define the coefficient-free reciprocal emitter

$$B_d(r) = \sum_{q \in \mathcal{Q}} \sum_{0 < |m| \leq dq/H} \psi\left(\frac{Hm}{dq}\right) \mathbf{1}_{m\bar{q} \equiv r \pmod{d}}, \quad r \in \mathbb{Z}/d\mathbb{Z}. \quad (2)$$

The V46 coefficient is kept separately as

$$c_d = \frac{\mu(d) \log d}{d}. \quad (3)$$

For a common physical variable u , put

$$K_d(u) = c_d \sum_{r \bmod d} B_d(r) e\left(\frac{ru}{d}\right), \quad K(u) = \sum_{d \in \mathcal{D}} K_d(u). \quad (4)$$

The cutoff is interpreted in the integer sense, so that $|m| \leq \lfloor dq/H \rfloor$. This convention is important when the dilation argument changes variables from m to kn .

Lemma 2.1 (dilation covariance). *Suppose $h \mid d$ and write $d = kh$. For every $a \in \mathbb{Z}/h\mathbb{Z}$,*

$$B_d(ka) = B_h(a). \quad (5)$$

Proof. In the d -row, the congruence $m\bar{q} \equiv ka \pmod{kh}$ is equivalent to $m = kn$ and $n\bar{q} \equiv a \pmod{h}$. Indeed, q is a unit modulo kh , so the first congruence forces $k \mid m$. The range becomes $|n| \leq hq/H$, and the smooth argument is unchanged:

$$\frac{H(kn)}{(kh)q} = \frac{Hn}{hq}.$$

Term-by-term comparison with (2) proves the identity. $\square$

Corollary 2.2 (zero axis). *If every $q \in \mathcal{Q}$ satisfies $q < H$, then $B_d(0) = 0$ for every d . Hence the reduced denominator $h = 1$ contributes no physical Gram energy.*

Proof. The congruence for $r = 0$ requires $d \mid m$. But every nonzero admissible m satisfies $|m| \leq dq/H < d$, which is impossible. $\square$

3 The cluster factorization theorem

Let

$$L = \text{lcm}\{d : d \in \mathcal{D}\}, \quad \mathcal{H}_* = \{h \geq 2 : h \mid d \text{ for some } d \in \mathcal{D}\}. \quad (6)$$

For $h \in \mathcal{H}_*$ define the Möbius-log tail

$$C_h = \sum_{\substack{d \in \mathcal{D} \\ h \mid d}} c_d. \quad (7)$$

Theorem 3.1 (reduced-denominator cluster factorization). *Assume the hypotheses of Lemma 2.1. On one complete period,*

$$\sum_{u=0}^{L-1} |K(u)|^2 = L \sum_{h \in \mathcal{H}_*} |C_h|^2 \sum_{\substack{a \bmod h \\ (a,h)=1}} |B_h(a)|^2. \quad (8)$$

The formula remains valid for complex-valued ψ after replacing the inner square by the Hermitian norm of the row.

Proof. Every nonzero rational frequency in the d -row has a unique reduced form a/h with $h \mid d$ and $(a, h) = 1$. Its representative in $\mathbb{Z}/d\mathbb{Z}$ is $r = (d/h)a$. Lemma 2.1 gives

$$c_d B_d((d/h)a) = c_d B_h(a).$$

Collecting all divisors that contain the same reduced frequency therefore gives

$$K(u) = \sum_{h \in \mathcal{H}_*} C_h \sum_{\substack{a \bmod h \\ (a,h)=1}} B_h(a) e(au/h),$$

where the zero-frequency term is absent by Corollary 2.2. Distinct reduced fractions have distinct frequencies modulo one. Summing their products over $u = 0, \dots, L-1$ gives zero unless the fractions agree, in which case it gives L . The surviving terms are exactly (8). $\square$

Corollary 3.2 (relation with the pairwise Gram). *The pairwise cross-divisor Gram is not the fundamental object once the coefficient is restored. It is the expansion of the cluster sum:*

$$L \sum_{d,e \in \mathcal{D}} c_d c_e \sum_{\substack{r \bmod d, s \bmod e \\ r/d \equiv s/e \pmod{1}}} B_d(r) B_e(s) = \text{right-hand side of (8)}. \quad (9)$$

In particular, the physical period L must be used consistently for every pair; replacing it by $\text{lcm}(d, e)$ inside a family sum loses the number of repeated periods.

Proof. Expand the square in (8), use Lemma 2.1, and apply complete-period orthogonality. The final sentence is the same identity with the common period written as a multiple of each pair period. $\square$

Table 1: Exact-rational row computation followed by numerical logarithm evaluation.

divisor family $\mathcal{D}$	L	physical cluster energy	direct-sum energy	ratio
$\{5,7,35\}$	35	35.0705402044	58.8092884914	0.5963435557
$\{3,5,7,105\}$	105	183.3803044582	151.3044743922	1.2119952513

4 Four-packet compatibility

The Gate-B route uses the complex polarization $a^{(j)} = \beta + i^j w$. The cluster theorem is linear in the row before taking the Hermitian square, so the polarization can be applied after the frequency reduction.

Proposition 4.1 (polarization commutes with clustering). *For any $\beta, w \in \mathbb{C}$,*

$$\frac{1}{4} \sum_{j=0}^3 i^j |\beta + i^j w|^2 = \beta \bar{w}. \quad (10)$$

If four physical kernels are formed from the rows $\beta + i^j w$, then each reduced-frequency cluster may be formed first and (10) applied afterwards. The resulting polarized scalar is not positive in general.

Proof. Expand the square. The sums of i^j and i^{2j} vanish, while $i^j \bar{i}^j = 1$. The remaining term is $\beta \bar{w}$. All maps used to form a cluster are linear, so the same calculation applies cluster by cluster. $\square$

Remark 4.2 (scope of the zero axis). The zero-axis statement concerns the additive frequency $r = 0$ in the reciprocal emitter. It does not delete the multiplicative principal character, a local Euler exception, or any other mode in a later character decomposition.

5 Exact finite certificate

The release certificate uses

$$\mathcal{Q} = \{11, 13, 17\}, \quad H = 40, \quad \psi(t) = \frac{1}{(1+t^2)^2}. \quad (11)$$

The profile is smooth and even, and all finite row entries are rational. The two divisor families were chosen to test opposite signs rather than to optimize a favorable outcome. Energies use the common family period L in both the physical and direct-sum columns.

For the first family, the active coefficient tail at the three nonzero reduced denominators is approximately

$$C_5 = -0.2203062093, \quad C_7 = -0.1764057910, \quad C_{35} = 0.1015813732.$$

The physical square sees these tails before squaring. The direct-sum column instead retains the three divisor diagonal energies separately. For the second family, the quotient structure involving 105 changes the tail signs enough to produce a ratio above one. The sign directions are exact: in the first family, the only nonzero cross terms pair a negative prime coefficient with the positive two-prime coefficient; in the second family all four coefficients are negative and at least one cross Gram is positive. The decimal ratios are numerical observations. This is a finite sign obstruction, not a lower bound for the actual V46 object.

The producer records exact rational row hashes, exact rational primitive cluster norms, exact rational Gram entries, and a formal polynomial in the variables $\log p$. The independent checker reconstructs those data without importing the producer module. Both normal and optimized Python executions pass, and a separate exact Gaussian-rational check verifies (10).

6 Route evaluation and limitations

Route A is not applicable: this is the analytic prime/twin-prime session rather than a dynamical spectral family. The Route-B structural evaluation is

emitter dilation covariance	PROVED_EXACT
reduced-denominator cluster factor	PROVED_EXACT
zero-axis scope	PROVED_EXACT
four-packet polarization	PROVED_EXACT_LINEAR_EXTENSION
finite cancellation/enhancement signs	PROVED_EXACT_FINITE_SIGN
finite energy ratios	NUMERICAL_OBSERVATION
universal favorable sign	REFUTED_SCOPED
literal V46 asymptotic cluster bound	OPEN
prime-shell reassembly	OPEN
arithmetic advance	NO
strict 1/400 endpoint	UNPAID

The exact theorem identifies the correct object for the next estimate, but it does not estimate the coefficient tails uniformly in the transition band $Y_0 < d \leq U$. It also does not perform the prime-only shell reassembly or attach a source-backed collective BDH theorem to the four polarized packets. The finite profile is one explicit Schwartz choice, not the proof of an asymptotic profile law. Accordingly, the release earns structural route advance only: no arithmetic L^2 credit, fixed-atom credit, strict 1/400 payment, or twin-prime conclusion is claimed.

7 Conclusion

TPC-214 turns the shared-frequency observation from TPC-213 into an exact reduced-denominator theorem. The common-source physical Gram is governed by Möbius-log tails C_h , not by independent divisor energies. This is the reusable structure for the next arithmetic step. The two finite certificates also settle the sign question at the right scope: nested coupling can cancel, but it can also enhance. The remaining theorem must therefore control these tails with the literal smooth profile, four-packet signs, zero axis, and prime shell all retained.

References

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